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1D Convolutional Autoencoder for ECG Signal Denoising and Classification

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ABSTRACT

INDEX TERMS CAE, Classification, CNN, CVD, Decoder, Denoising, DWT, ECG, EMD, Encoder, MLII, SWT

I. INTRODUCTION

Cardiovascular disease (CVD) is one of the leading causes of death globally, and it involves all disorders that affect the heart and blood circulation or vessels [1]. For this reason, it is necessary to enhance all detection techniques to be able to early detect any irregularities to avoid fatal stages.

One of the most important techniques is the electrocardiogram (ECG), which is a test that can be used to monitor a patient's heart rhythm and electrical activity. The electrical signals produced by the heart are measured using electrodes attached to the patient's skin. The heart activity is recorded to be analyzed by specialists to investigate symptoms of possible heart problems or any abnormalities.

Despite the usefulness of ECG, the recorded signals are prone to various noise sources, which may distort the heart's signal, leading to misdiagnosis. There are many reasons for having noise during measurement.

The following are the main causes of noise:

- Noises caused by electrode movement (EM), which usually have a large amplitude.
- High-Frequency Noises caused by muscle artifact (MA), which is a muscle noise caused by involuntary muscle movements like shivering, tension, or tremors.
- Low-Frequency Noises caused by baseline wander (BW), mainly caused by human breathing /respiration [2].

The mentioned noise types can mask important diagnostic features in the signal, so effectively removing these noises without altering the essential morphology of the ECG signal remains a significant and ongoing challenge[3].

Thus, various denoising techniques have been developed to enhance the quality of ECG signals. This study aims to implement and train a neural network-based filtering/denoising technique using a convolutional autoencoder(CAE).

This paper is structured as follows:

- **Section II** which covers the necessary background information, starting from
- **Section III** which outlines
- **Section IV** which shows
- **Section VI** which compares
- Lastly, **Section VII** explains the major challenges and research gaps, showing that denoising the ECG signal is a very active topic in research, and there are multiple future directions that need to be explored.

II. BACKGROUND AND LITERATURE REVIEW

A. ECG SIGNAL AND ITS CHARACTERISTICS

A full ECG is obtained via 10 electrodes, that captures 12 leads (signals). Each lead identifies the electrical activity from a different angle/perspective in order to get a complete picture of the heart. However, a single lead can offer adequate information for a quick and initial assessment of the patient (Modified Limb Lead 2 (MLII) is used in this paper, which provides a clear QRS complex and visible P-Waves making it suitable for heartbeat detection). One ECG cycle consists of five waves: P, Q, R, S, and T that map different phases of the heart's activities.

The P-wave is the first positive deflection on the ECG, The QRS represents the simultaneous activation of the right and left ventricles, although most of the QRS waveform is derived

from the larger left ventricular musculature and the T-wave should be concordant with the QRS complex. P-R interval measured from the beginning of the P-wave to the first deflection of the QRS complex and Q-T interval measured from the first deflection of QRS complex to end of T-wave at the isoelectric line as shown in Fig. 1 [4].

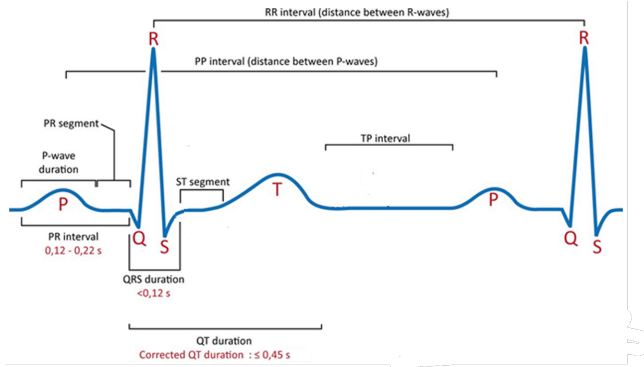


FIGURE 1. ECG Waveform [4]

B. TRADITIONAL ECG SIGNAL DENOISING TECHNIQUES

There are several techniques that addressed ECG signal denoising, including **time-frequency analysis techniques** such as wavelet transforms like discrete wavelet transform(DWT), and Stationary wavelet transform (SWT), and techniques like **empirical mode decomposition (EMD)** and **filtering-based techniques**, which are widely used due to their effectiveness in ECG denoising [5]. These techniques have shown significant results in separating noise from the ECG signal. For instance, wavelet transform-based techniques are efficient at isolating components in both time and frequency domains to remove baseline drift or high-frequency noise. On the other hand, adaptive filters can flexibly adjust to variable noise conditions, while EMD decomposes signals into intrinsic mode functions (IMF) to isolate noise-dominant components [6], [7]. Despite this, such classical techniques are not sufficient as they face notable challenges and limitations that struggle with the critical morphological characteristics of the ECG signal. For instance, muscle artifact noise often overlaps with the frequency bands of P-waves, T-waves, and QRS complexes (important features of ECG signals), leading to signal distortion or loss of clinically relevant information. These overlaps make it difficult for customized filters to distinguish between noise and meaningful signal content, especially under low SNR conditions [8], [9].

C. DATA-DRIVEN ECG SIGNAL DENOISING TECHNIQUES

D. 1D CONVOLUTION

E. AUTOENCODER

III. METHODOLOGY

A. DATASET

B. ARCHITECTURE

IV. EVALUATION

A. EVALUATION METRICS

V. RESULTS

VI. ABLATION STUDIES

VII. CHALLENGES AND FUTURE DIRECTIONS

VIII. CONCLUSION

REFERENCES

- [1] World Health Organization, "Cardiovascular diseases (cvds)," [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)), 2026.
- [2] M. T. Almalchy, V. Ciobanu, and N. Popescu, "Noise removal from ecg signal based on filtering techniques," in 2019 22nd International Conference on Control Systems and Computer Science (CSCS), May 2019, pp. 176–181.
- [3] A. Ghandehari, J. A. Tavares-Negrete, X. Pei, J. Rajendran, S. Chakoma, and R. Esfandyarpour, "Optimizing nfc-based wearable sensors for arterial pulse monitoring: A comparative study of sampling rates and machine learning models," in 2024 IEEE 20th International Conference on Body Sensor Networks (BSN), Oct. 2024, pp. 1–4.
- [4] M. Hammad, A. Maher, K. Wang, F. Jiang, and M. Amrani, "Detection of abnormal heart conditions based on characteristics of ecg signals," *Measurement*, vol. 125, pp. 634–644, 2018.
- [5] H. Peng, X. Chang, Z. Yao, D. Shi, and Y. Chen, "A deep learning framework for ecg denoising and classification," *Biomedical Signal Processing and Control*, vol. 94, p. 106441, 2024.
- [6] M.-B. Hossain, S. K. Bashar, J. Lazaro, N. Reljin, Y. Noh, and K. H. Chon, "A robust ecg denoising technique using variable frequency complex demodulation," *Computer methods and programs in biomedicine*, vol. 200, p. 105856, 2021.
- [7] C. Li, Y. Wu, H. Lin, J. Li, F. Zhang, and Y. Yang, "Ecg denoising method based on an improved vmd algorithm," *IEEE Sensors Journal*, vol. 22, no. 23, pp. 22 725–22 733, 2022.
- [8] G. D. Clifford, F. Azuaje, P. Mcsharry et al., "Ecg statistics, noise, artifacts, and missing data," *Advanced methods and tools for ECG data analysis*, vol. 6, no. 1, p. 18, 2006.
- [9] J. Lim, D. Han, M. P. S. Nejad, and K. H. Chon, "Ecg classification via integration of adaptive beat segmentation and relative heart rate with deep learning networks," *Computers in biology and medicine*, vol. 181, p. 109062, 2024.

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